

Intelligence

Lecture 4

Theories of Intelligence

Dr Steven Poulter

Key Themes

- What is g ? Finding structure in the data
- Spearman's 2-factor theory of intelligence
- Thurstone's primary mental abilities
- Carroll's 'three stratum model' of human cognitive ability
- Cattell's fluid (g_f) and crystallized intelligence (g_c)
- Psychometric vs. Cognitive approaches
- Gardner's theory of multiple intelligences



Glossary of Key Terms

Factor Analysis

A statistical technique used to identify clusters of related variables. It reveals underlying “latent factors” (e.g., intelligence, personality traits or depression) that explain correlations among tests.

Correlation

A measure of how strongly two variables relate. Positive correlations underpin the discovery of *g* and broader mental ability groupings.

***g* (General Intelligence)**

The general mental ability underlying performance across all cognitive tasks. Proposed by Spearman to explain why test scores are positively correlated.

Specific Abilities (*s*)

Task-specific skills unique to particular tests (e.g., memory for digits, spatial rotation). In Spearman’s model, each test reflects *g* + a specific factor.

Spearman’s Two-Factor Theory

Intelligence consists of:

- g*** — a general ability shared by all tests, and
- s*** — abilities unique to each task.

Glossary of Key Terms

Primary Mental Abilities (Thurstone)

Seven broad, separable abilities (e.g., verbal comprehension, spatial, numerical). A challenge to Spearman, though still positively correlated.

Positive Manifold

The consistent finding that all cognitive tests positively correlate with each other. The main empirical basis for g .

Three-Stratum Model (Carroll)

A hierarchical model summarising decades of factor-analytic data:

- **Stratum III:** g
- **Stratum II:** broad abilities (e.g., memory, spatial, verbal) - *cognitive domains*
- **Stratum I:** narrow, task-specific abilities – *cognitive tasks*

Fluid Intelligence (gf)

Ability to solve novel problems, reason abstractly, and detect patterns. Less dependent on prior knowledge; measured well by Raven's Progressive Matrices.

Crystallized Intelligence (gc)

Knowledge-based skills accumulated through experience and education (e.g., vocabulary, general knowledge). Increases across adulthood.

Glossary of Key Terms

Psychometric Approach

Studying intelligence through test scores and their statistical relationships. Focus: *What patterns are in the data?*

Cognitive Approach

Examines the mental processes and biological mechanisms underlying performance. Focus: *How does the mind solve tasks?*

Elementary Cognitive Tasks (ECTs)

Simple tasks measuring speed and efficiency of basic processes (e.g., reaction time, inspection time). Moderately correlated with *g*.

Reaction Time (RT)

Speed of responding to a stimulus. Lower RT and lower variability tend to correlate with higher intelligence.

Inspection Time (IT)

The minimum time needed to discriminate between stimuli. Shorter IT is associated with higher *gf*.

Multiple Intelligences (Gardner)

A non-psychometric theory proposing several independent intelligences (e.g., linguistic, musical, spatial, interpersonal). Based on neuropsychology, prodigies, cultural evidence—not factor analysis.

Glossary of Key Terms

Latent Variable

An unobserved construct inferred from measurable indicators, such as g being inferred from correlations across tests.

Eigenvalue

A statistic in factor analysis indicating how much variance a factor explains. Factors with eigenvalues >1 are often retained.

Communality

The proportion of variance in each variable explained by the retained factors.

Rotated Component Matrix

A table showing how strongly each variable loads on each factor after rotation, used to interpret and label the factors.

Symbol Systems (Gardner)

Representational tools associated with each intelligence (language, maps, musical notation), used as part of Gardner's evidence for modularity.

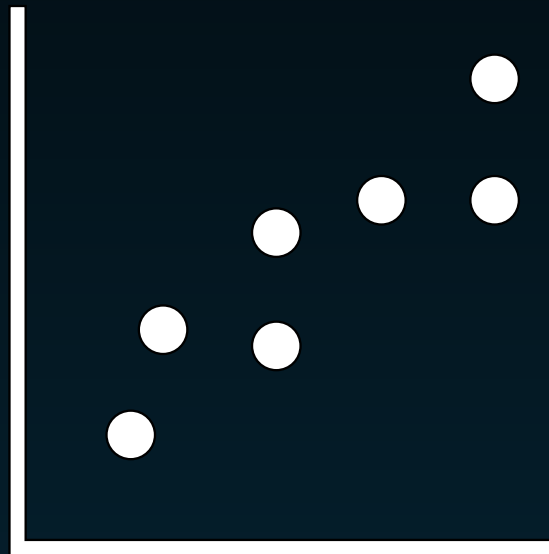
Neuropsychological Dissociation

Cases where one cognitive ability is impaired while another is preserved, used to argue for separability of abilities.

Uncovering g

Uncovering structure in your data....

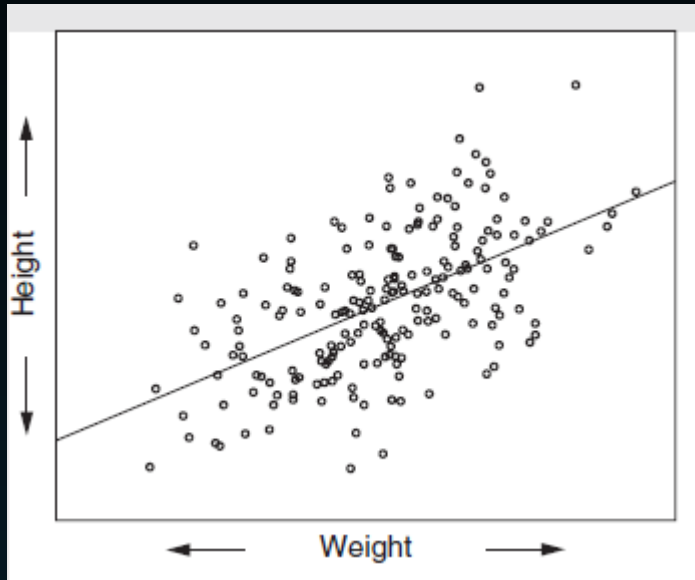
- Correlations are the basis of factor analysis
- Task of factor analysis is to group variables that are most related, or *correlated*



Example Correlations

1

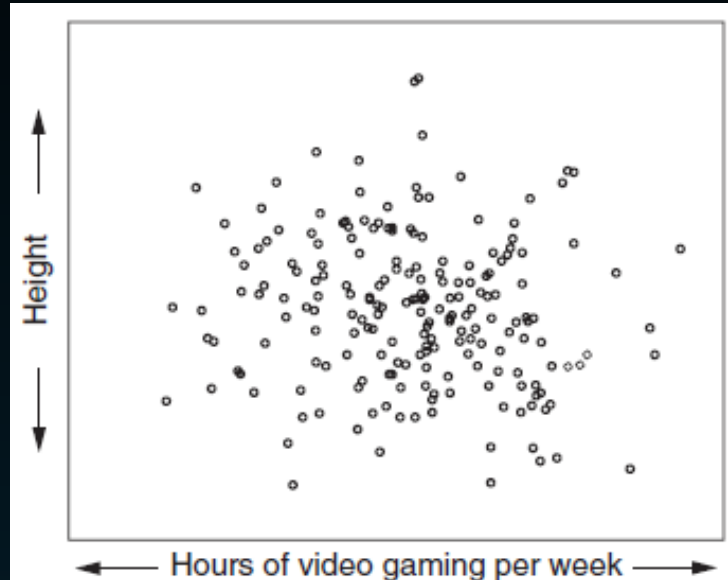
Positive Correlation



Height increases
with weight

0

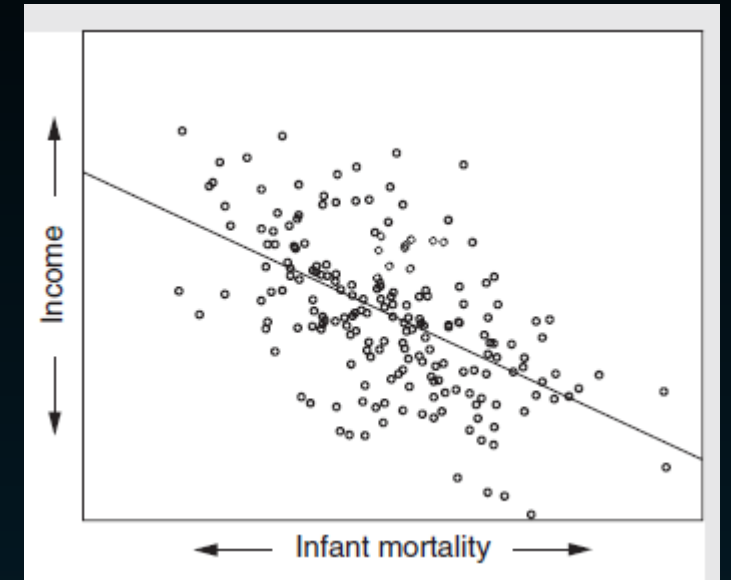
No Correlation



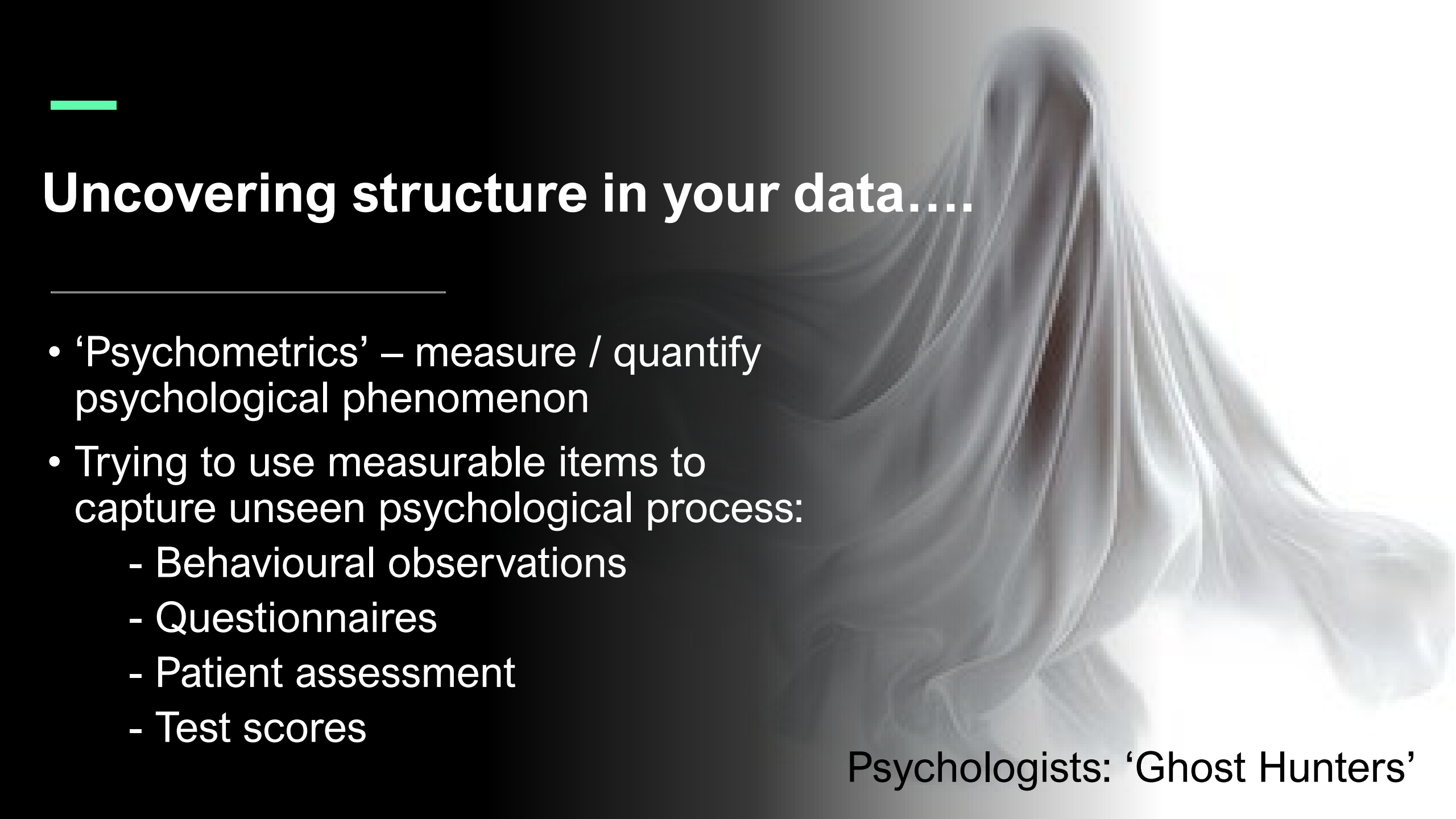
No correlation
between height and
hours spent gaming

-1

Negative Correlation



As family income
goes up infant
mortality goes down



Uncovering structure in your data....

- ‘Psychometrics’ – measure / quantify psychological phenomenon
- Trying to use measurable items to capture unseen psychological process:
 - Behavioural observations
 - Questionnaires
 - Patient assessment
 - Test scores

Psychologists: ‘Ghost Hunters’

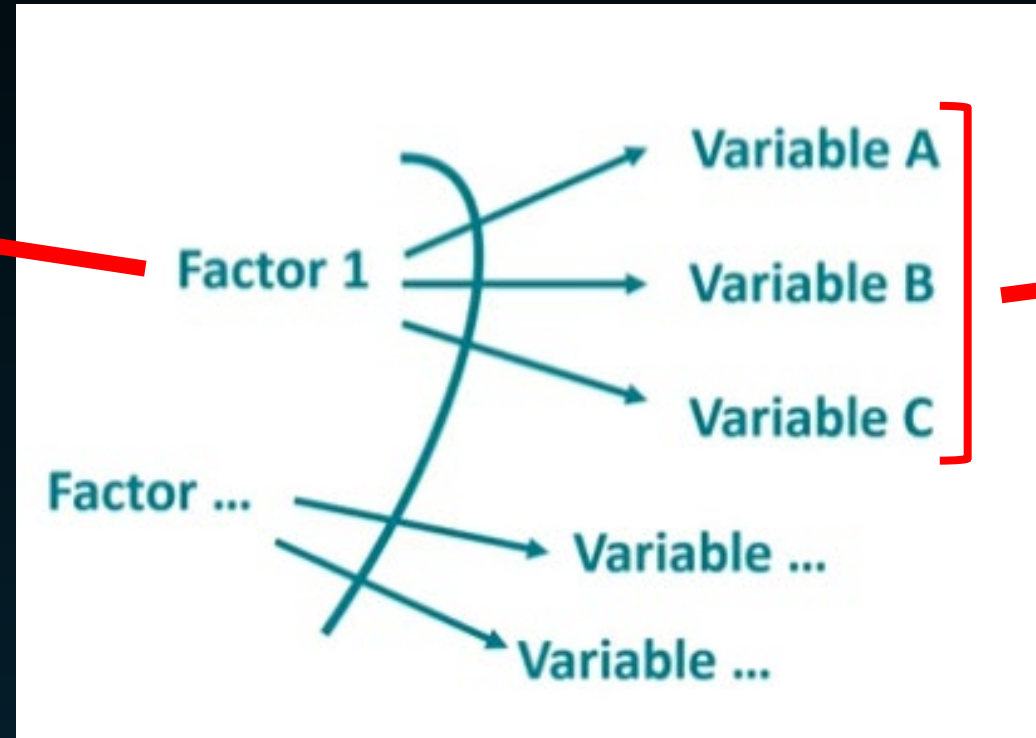
Uncovering structure in your data....

- To measure constructs like depression or intelligence - need to find items that relate closely to these constructs (validity)
- Items need to be consistent across samples and time
- If items reliably measuring the same process, e.g. depression or intelligence, they should all be related to each other
- ...because we often want to create a single score for each person and trust that score is measuring e.g. high depression
- How can we do that with statistical approaches?

Uncovering structure in your data: an example

Task of factor analysis is to group variables that are most related, or *correlated*

Hidden /
Latent
variable



Observed
variables

Uncovering structure in your data: an example

Can different personality traits be grouped into personality types?

Personality traits

Sociable
Hard-working
Warm-hearted
Outgoing
Dutiful
Helpful

Personality types

Conscientiousness
Extraversion
Agreeableness

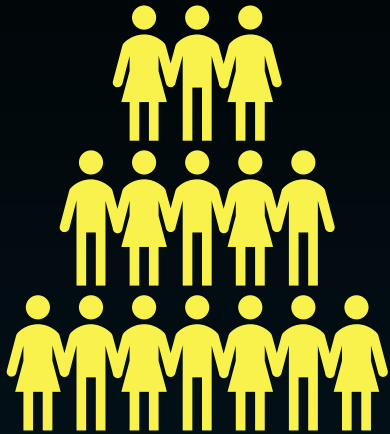
To find out you create a Questionnaire

Please indicate how strongly the following characteristics apply to you. I am...

	applies	rather applies	partially applies	rather not applies	does not apply
outgoing	☐	☐	☐	☐	☐
sociable	☐	☐	☐	☐	☐
hard-working	☐	☐	☐	☐	☐
dutiful	☐	☐	☐	☐	☐
warm-hearted	☐	☐	☐	☐	☐
helpful	☐	☐	☐	☐	☐

Do some of the traits correlate strongly with each other – can they be clustered into a ***factor***?

Interview 20 people and put results in Excel



AutoSave Off PCA_Example.xlsx

File Home Insert Page Layout Formulas Data Review View Automate Developer Help Acrobat

Clipboard Font Alignment Number Styles

Calibri 11 B I U Wrap Text Merge & Center General Conditional Formatting Format as Table Normal Bad Good Neutral

	A	B	C	D	E	F	G
1	person	outgoing	sociable	hard-working	dutiful	warm-hearted	helpful
2	1	4	5	4	5	4	3
3	2	1	2	5	4	1	4
4	3	1	2	2	3	4	2
5	4	4	5	1	2	2	1
6	5	5	5	5	3	4	3
7	6	1	2	2	3	3	3
8	7	3	3	4	4	5	5
9	8	2	2	5	4	1	1
10	9	2	1	1	1	1	4
11	10	1	4	4	4	4	1
12	11	2	3	5	4	2	3
13	12	4	5	1	2	3	2
14	13	2	3	2	3	4	5
15	14	3	2	3	2	4	5
16	15	2	2	3	5	2	1
17	16	2	4	5	3	5	4
18	17	4	2	3	2	1	1
19	18	3	2	5	1	2	2
20	19	5	4	1	5	2	1
21	20	2	1	1	1	5	5

Uncovering structure in your data: an example

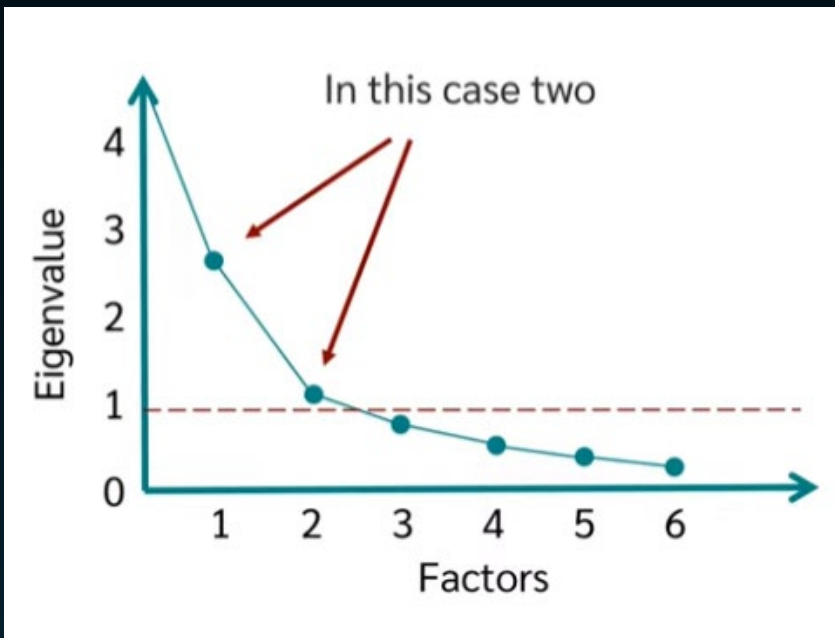
Please indicate how strongly the following characteristics apply to you. I am...

	applies	rather applies	partially applies	rather not applies	does not apply	
outgoing	☐	☐	☐	☐	☐	} Extraversion
sociable	☐	☐	☐	☐	☐	
hard-working	☐	☐	☐	☐	☐	} Conscientiousness
dutiful	☐	☐	☐	☐	☐	
warm-hearted	☐	☐	☐	☐	☐	} Agreeableness
helpful	☐	☐	☐	☐	☐	

Our prediction before running factor analysis might be that the trait variables can be clustered into 3 factors

Uncovering structure in your data: an example

- We calculate factor analysis on all 6 metric variables (outgoing, sociable, hard-working, dutiful, warm-hearted, helpful)
- We also have to choose the number of factors....but how?



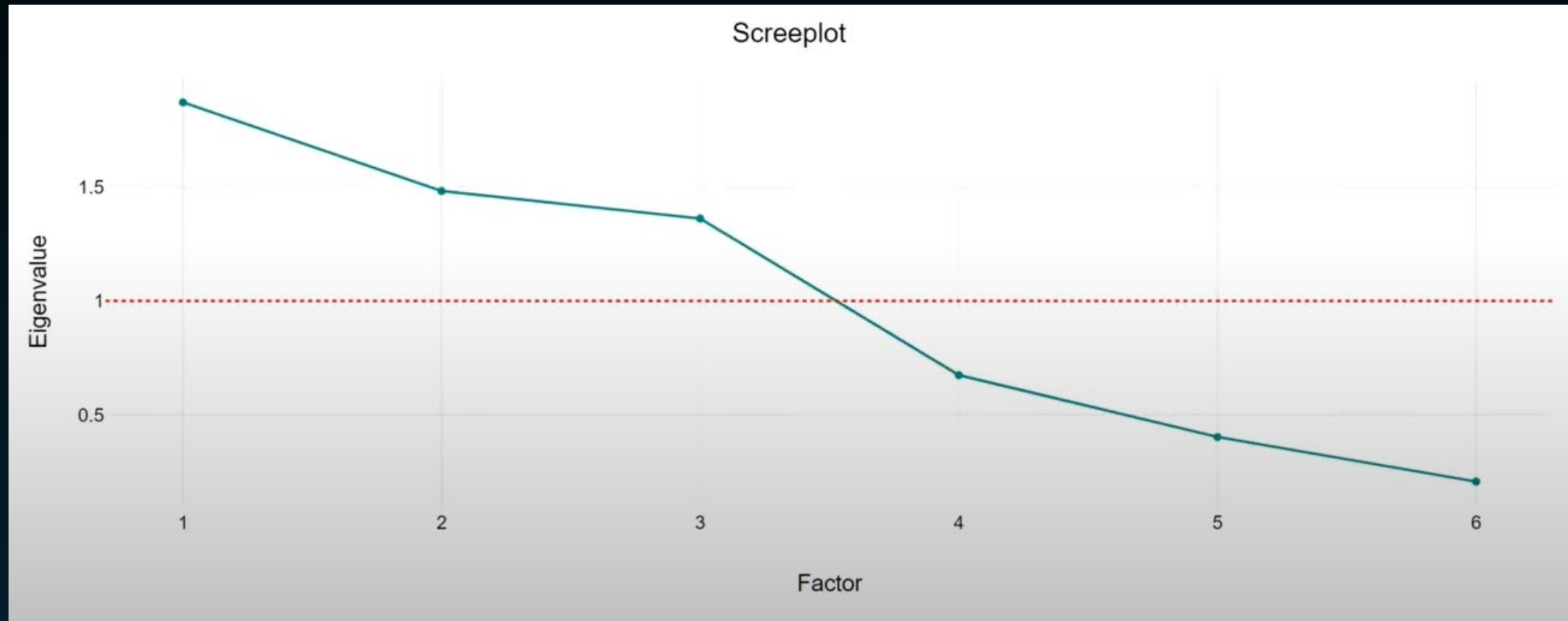
Factor analysis does not give you a clear answer

One method is to sort eigenvalues by size, plot on graph and see how many eigenvalues are greater than 1.

In this case you would select **2 factors**.

Uncovering structure in your data: an example

- For our example, 3 factors have an eigenvalue greater than 1, so we can select **3 factors**.



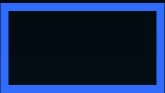
Uncovering structure in your data: an example

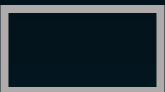
- We can look at the generated *correlation matrix*

Correlation matrix

Copy Word Copy Excel

	outgoing	sociable	hard-working	dutiful	warm-hearted	helpful
outgoing	1	0.58	-0.13	-0.02	-0.04	-0.22
sociable	0.58	1	0.07	0.33	0.27	-0.26
hard-working	-0.13	0.07	1	0.34	0	0.03
dutiful	-0.02	0.33	0.34	1	0	-0.25
warm-hearted	-0.04	0.27	0	0	1	0.52
helpful	-0.22	-0.26	0.03	-0.25	0.52	1


Strong


Weak

Uncovering structure in your data....

- We can also rank factors by size according to ***explained variance*** (left table) and ***communalities*** (how much variance explained by our 3 factors)

Explained total variance			
Copy Word  Copy Excel  			
Component	Total	% of variance	Accumulated %
1	1.87	31.21	31.21
2	1.48	24.71	55.93
3	1.36	22.7	78.62
4	0.67	11.23	89.86
5	0.4	6.7	96.56
6	0.21	3.44	100

Communalities	
Copy Word  Copy Excel  	
	Extraction
outgoing	0.77
sociable	0.88
hard-working	0.67
dutiful	0.72
warm-hearted	0.86
helpful	0.81

Rotated Component Matrix gives us the loadings of the factors (row) on the variables (column) – this rotation clusters the variables so that certain variables load as high as possible (and other variables load as low as possible) on each factor

Rotated Component Matrix (Varimax)

Copy Word  Copy Excel  

	Component		
	1	2	3
outgoing	0.84	-0.14	-0.23
sociable	0.9	0.07	0.25
hard-working	-0.13	0.05	0.8
dutiful	0.22	-0.16	0.81
warm-hearted	0.23	0.89	0.09
helpful	-0.3	0.84	-0.11

Variable 1

Variable 2

Variable 3

Rotated Component Matrix gives us the loadings of the factors (row) on the variables (column) – this rotation clusters the variables so that certain variables load as high as possible (and other variables load as low as possible) on each factor

Rotated Component Matrix (Varimax)

Copy Word  Copy Excel  

	Component		
	1	2	3
outgoing	0.84	-0.14	-0.23
sociable	0.9	0.07	0.25
hard-working	-0.13	0.05	0.8
dutiful	0.22	-0.16	0.81
warm-hearted	0.23	0.89	0.09
helpful	-0.3	0.84	-0.11

Extraversion

Note: Factor analysis does not tell us how to name the factors. That is our task.

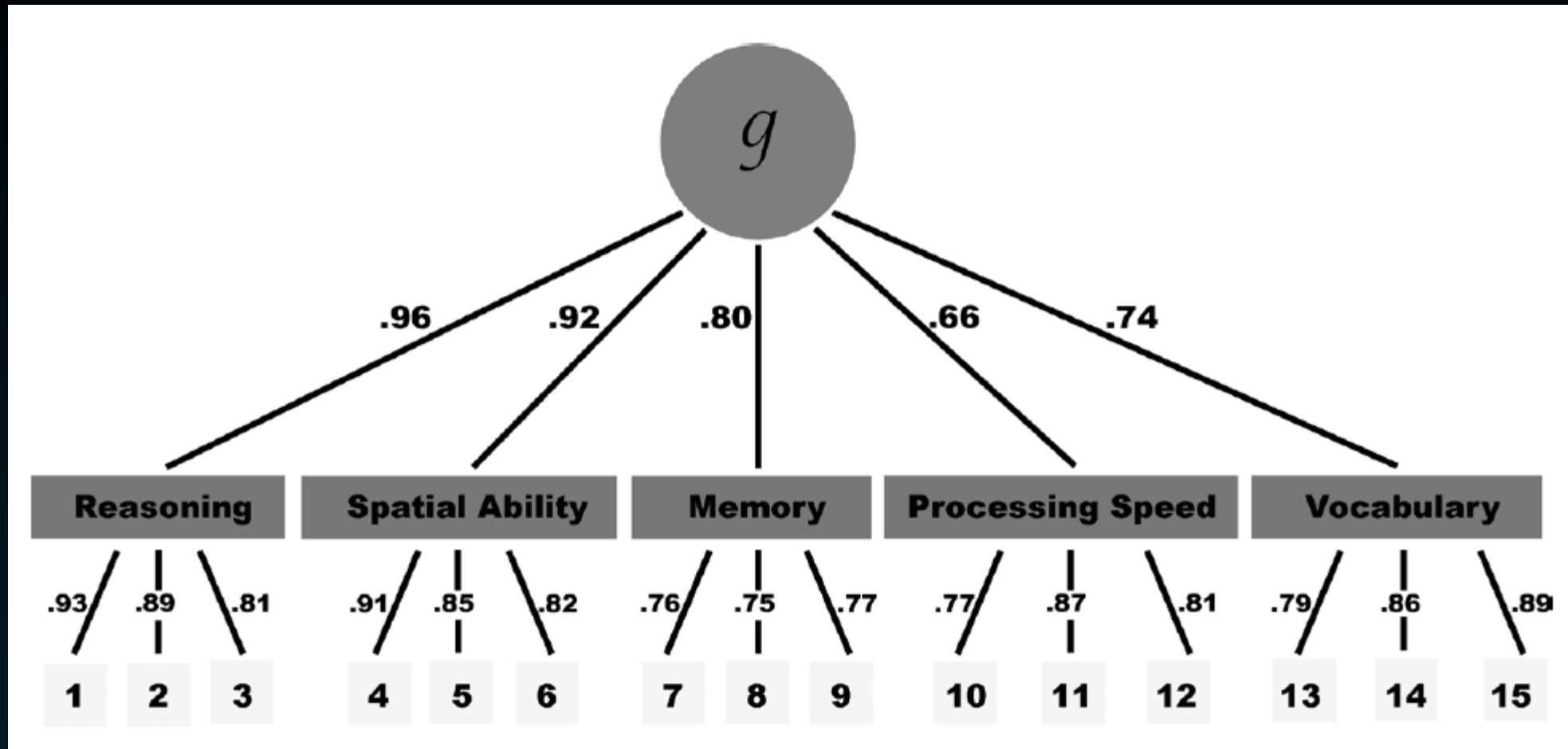
Conscientiousness

Agreeableness

SUMMARY Uncovering structure in your data....

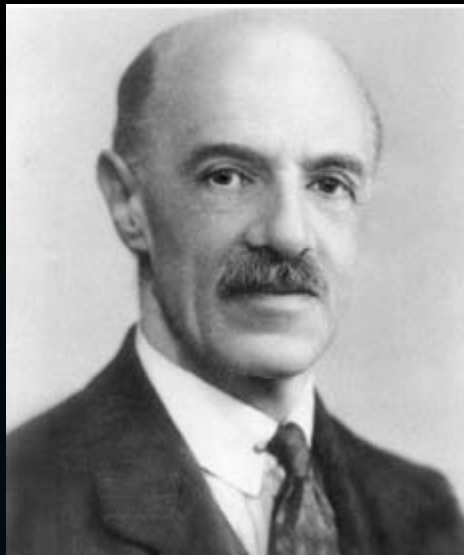
Factor analysis - more complex method of defining an operational definition – not using a single variable (but several) to define the unobservable factor, e.g. *g*-factor or general cognitive ability

The structure of mental abilities



The **g -factor** is common to all mental tests

Spearman's g



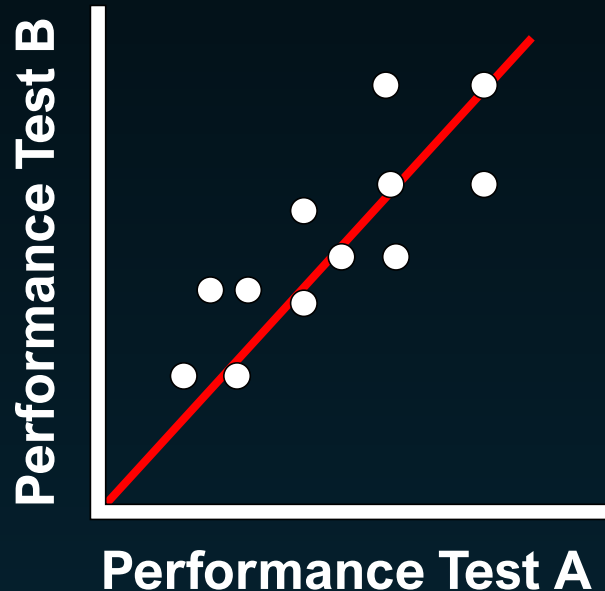
Charles Spearman
1863-1945

"GENERAL INTELLIGENCE," OBJECTIVELY DETERMINED AND MEASURED.

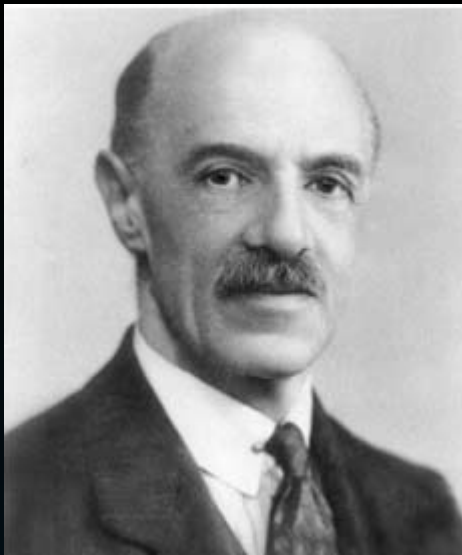
By C. SPEARMAN.

The American Journal of Psychology, Vol. 15, No. 2. (Apr., 1904), pp. 201-292.

1904 Spearman observed children's performance ratings, across seemingly unrelated subjects, were positively correlated



	Classics	French	English	Math	Pitch	Music
Classics	—					
French	.83	—				
English	.78	.67	—			
Math	.70	.67	.64	—		
Pitch discrimination	.66	.65	.54	.45	—	
Music	.63	.57	.51	.51	.40	—
<i>g</i>	.958	.882	.803	.750	.673	.646



Charles Spearman
1863-1945

**“GENERAL INTELLIGENCE,” OBJECTIVELY
DETERMINED AND MEASURED.**

By C. SPEARMAN.

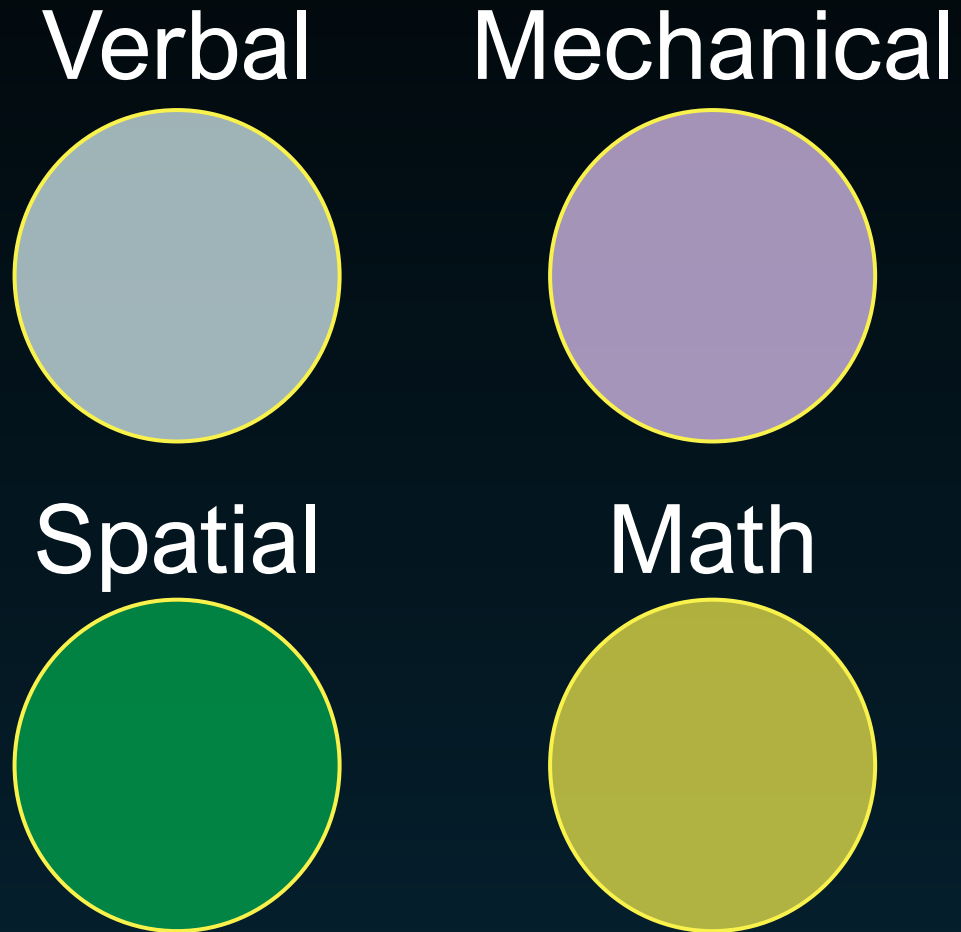
The American Journal of Psychology, Vol. 15, No. 2. (Apr., 1904), pp. 201-292.

- 1904 Seminal paper on factor analysis of intelligence
- Followed pioneering work by Galton using *correlations*
- Developed the earliest version of *factor analysis*
- Coined the term ***g*** or the g-factor
- Developed his 2-factor theory of intelligence

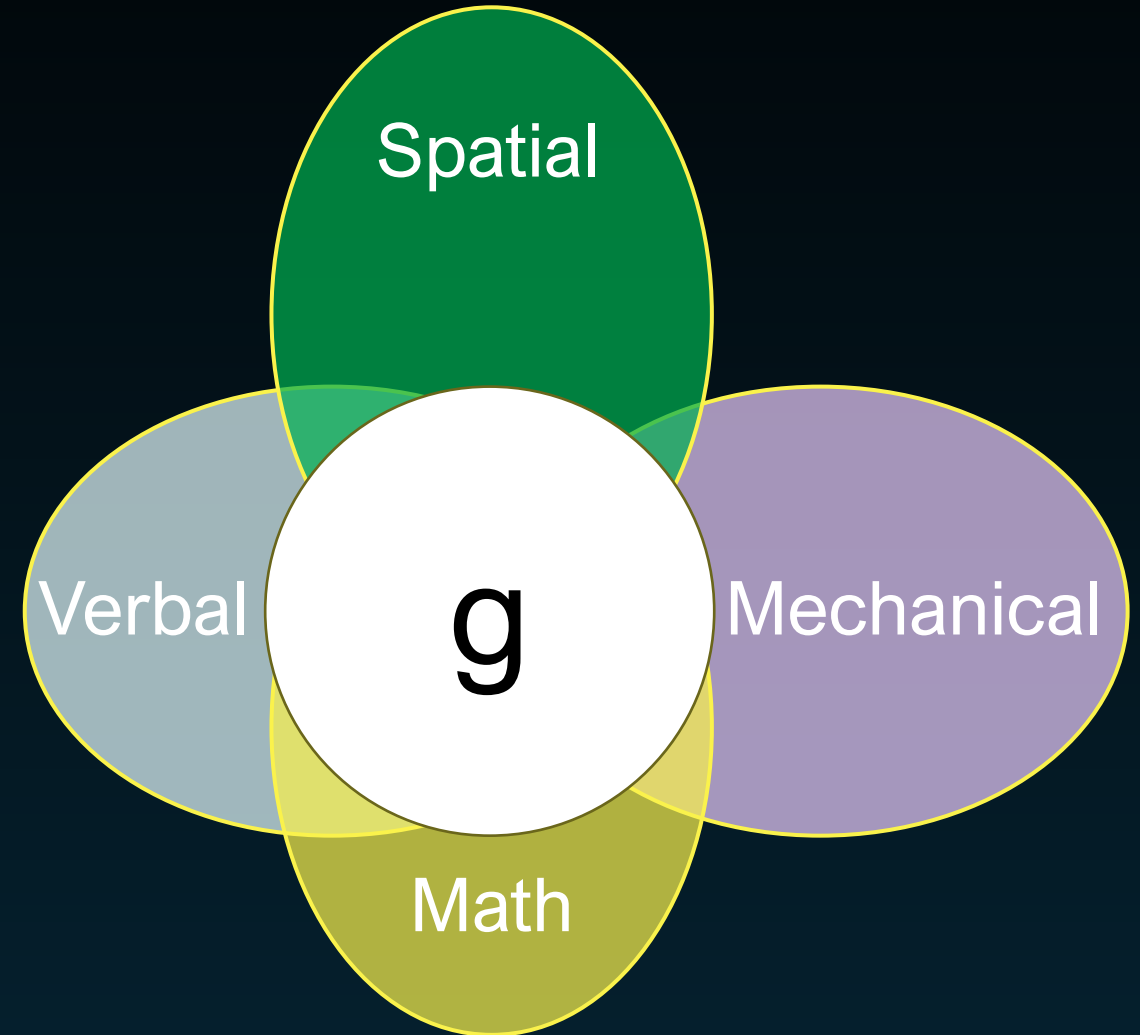
‘Under certain conditions the score of a person at a mental test can be divided into two factors, one of which is always the same in all tests, whereas the other varies from one test to another; the former is called the general factor or ***g, while the other is called the specific factor’***

Spearman's 2-factor theory of intelligence

Specific Intelligences



General Intelligence (g)

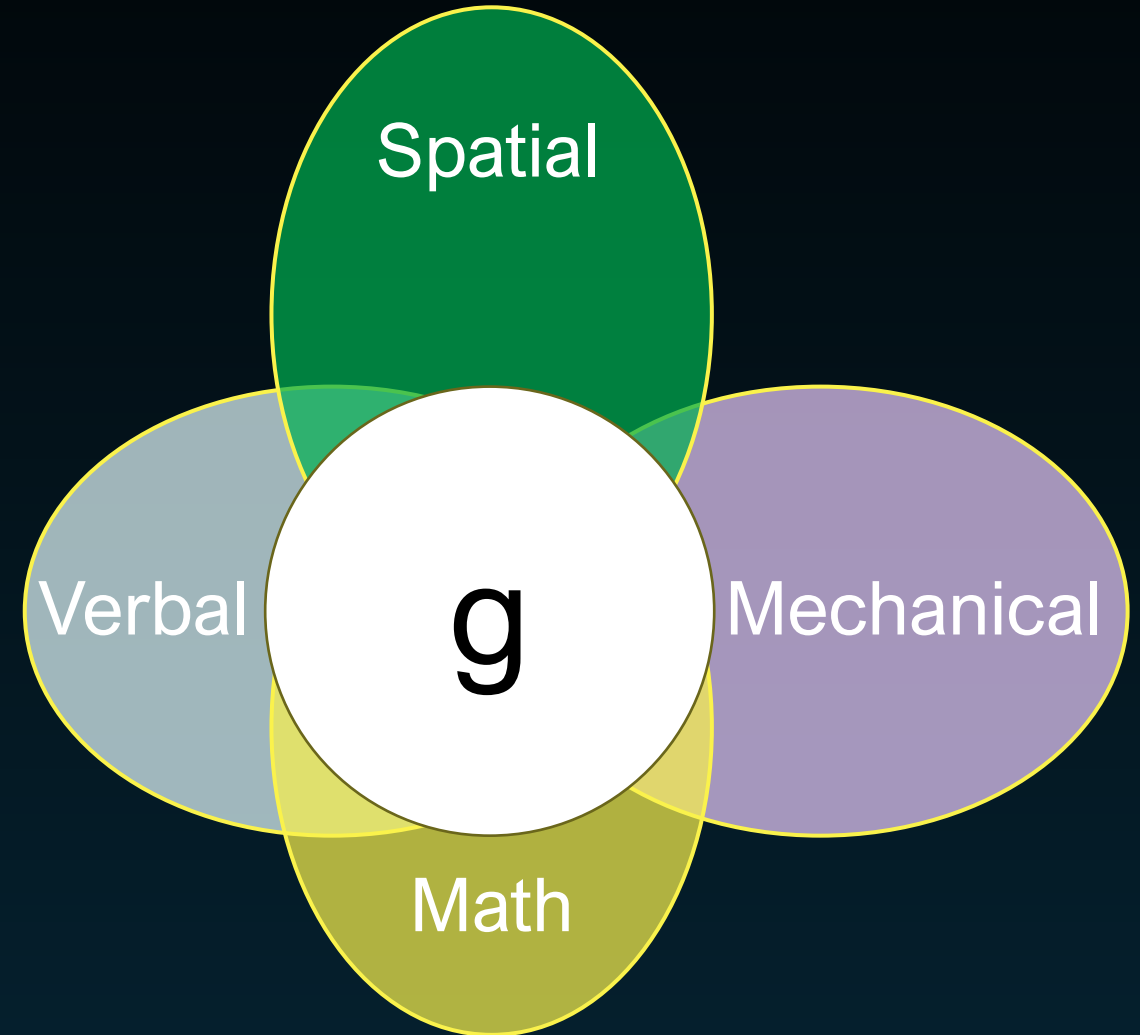


Spearman's 2-factor theory of intelligence

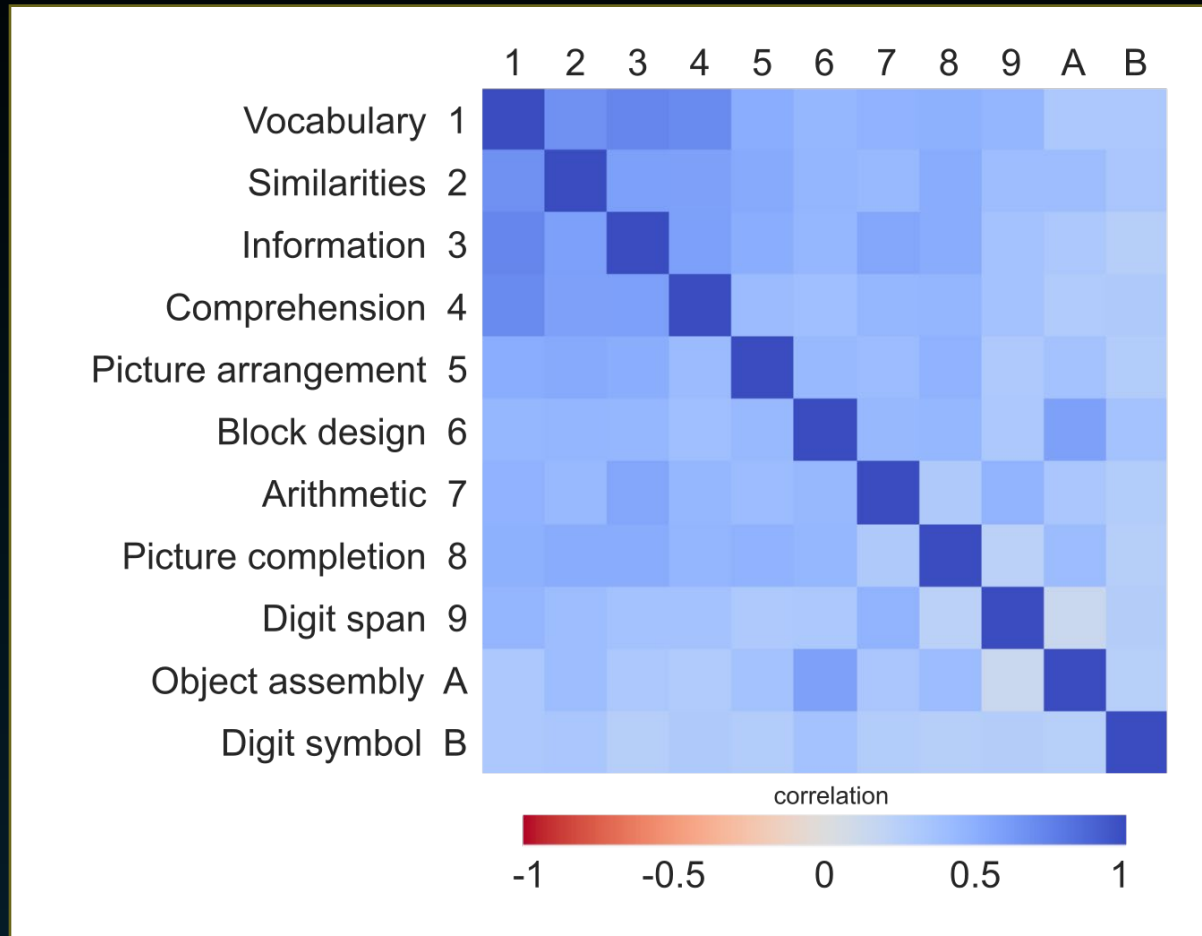
g (= general abilities)

- Underlying all the positive correlations. g is the intelligence required for performance of intelligence tests of all types.
- The mental energy that underlies specific factors of intelligence.
- A person's mathematical ability is not only affected by one's specific ability to perform a mathematical task but is also largely determined by that person's general intelligence.

General Intelligence (g)



Positive correlation effect – consistently replicated



	V	S	I	C	PA	BD	A	PC	DSp	OA	DS
V	–										
S	.67	–									
I	.72	.59	–								
C	.70	.58	.59	–							
PA	.51	.53	.50	.42	–						
BD	.45	.46	.45	.39	.43	–					
A	.48	.43	.55	.45	.41	.44	–				
PC	.49	.52	.52	.46	.48	.45	.30	–			
DSp	.46	.40	.36	.36	.31	.32	.47	.23	–		
OA	.32	.40	.32	.29	.36	.58	.33	.41	.14	–	
DS	.32	.33	.26	.30	.28	.36	.28	.26	.27	.25	–
<i>g</i>	.83	.80	.80	.75	.70	.70	.68	.68	.56	.56	.48

Data adapted from Chabris 2007, who credits Deary 2000

Thurstone's primary
mental abilities

Challenge to Spearman's unitary g -factor



Louis Leon Thurstone
(1887-1955)

- 1938: Thurstone applied a different statistical method – involved weighing the observed variables (e.g. scores on different ability subtests) differently
- This yielded not one but a set of “primary mental abilities”
- Most tests were positively correlated **BUT** scores on verbal tests were more highly correlated with other scores on other verbal tests than with scores on spatial tests

Challenge to Spearman's unitary g -factor



Louis Leon Thurstone
(1887-1955)

 Verbal domain

 Spatial domain

- Most tests were positively correlated **BUT** scores on verbal tests were more highly correlated with other scores on other verbal tests than with scores on spatial tests

WAIS subtests	Information (verbal)	Vocabulary (verbal)	Block design (spatial)	Object assembly (spatial)
Information (verbal)	-			
Vocabulary (verbal)	0.83	-		
Block design (spatial)	0.47	0.47	-	
Object assembly (spatial)	0.49	0.49	0.69	-

Thurstone's primary mental abilities



Louis Leon Thurstone
(1887-1955)

- Most tests were positively correlated **BUT** scores on verbal tests were more highly correlated with other scores on other verbal tests than with scores on spatial tests
 - Individuals have strengths and relative weaknesses in some cognitive domains
 - This might suggest that there is no such thing as “g”
 - **Clusters of correlations** suggest *primary mental abilities*.
(Note: We are not talking about specific abilities (“s”) here).

Thurstone's 7 primary mental abilities



Louis Leon Thurstone
(1887-1955)

- Thurstone, suggested that there were **seven** separable human mental abilities.

Mental Ability	Description
Word Fluency	Ability to use words quickly and fluency in performing such tasks as rhyming, solving anagrams, and doing crossword puzzles.
Verbal Comprehension	Ability to understand the meaning of words, concepts, and ideas.
Numerical Ability	Ability to use numbers to quickly compute answers to problems.
Spatial Visualization	Ability to visualize and manipulate patterns and forms in space.
Perceptual Speed	Ability to grasp perceptual details quickly and accurately and to determine similarities and differences between stimuli.
Memory	Ability to recall information such as lists or words, mathematical formulas, and definitions.
Inductive Reasoning	Ability to derive general rules and principles from the presented information.

Thurstone's 7 primary mental abilities



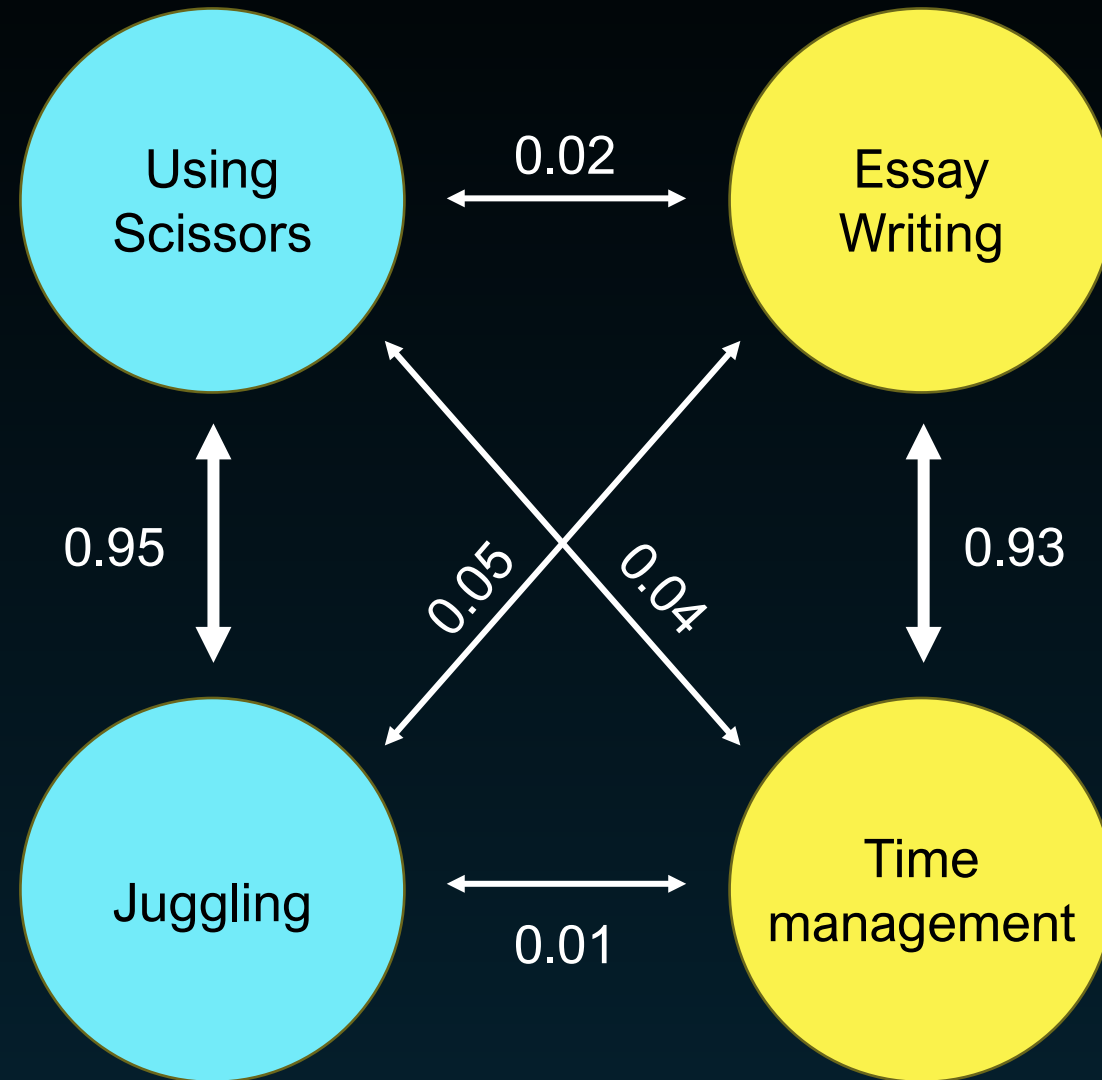
Louis Leon Thurstone
(1887-1955)

- Thurstone, suggested that there were **seven** separable human mental abilities.

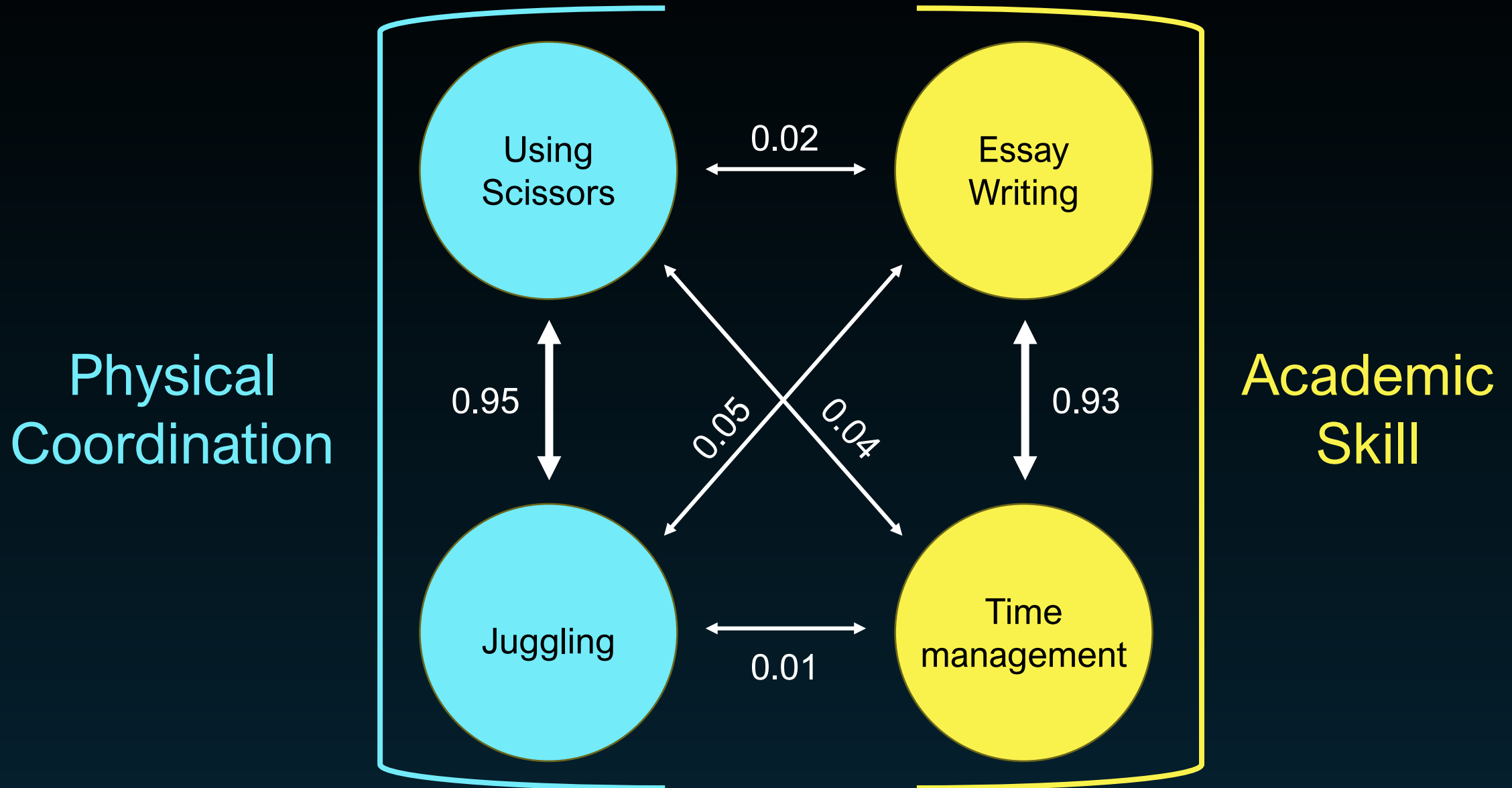
Thurstone de-emphasised 'g' and suggested instead that there were a number of stable and independent mental abilities, which were not general like *g* – e.g. a person might have weak verbal ability and strong numerical ability.

He suggested each person possesses different levels of these seven factors, and these levels do not depend on one another (however the seven were positively correlated even in his own dataset!!)

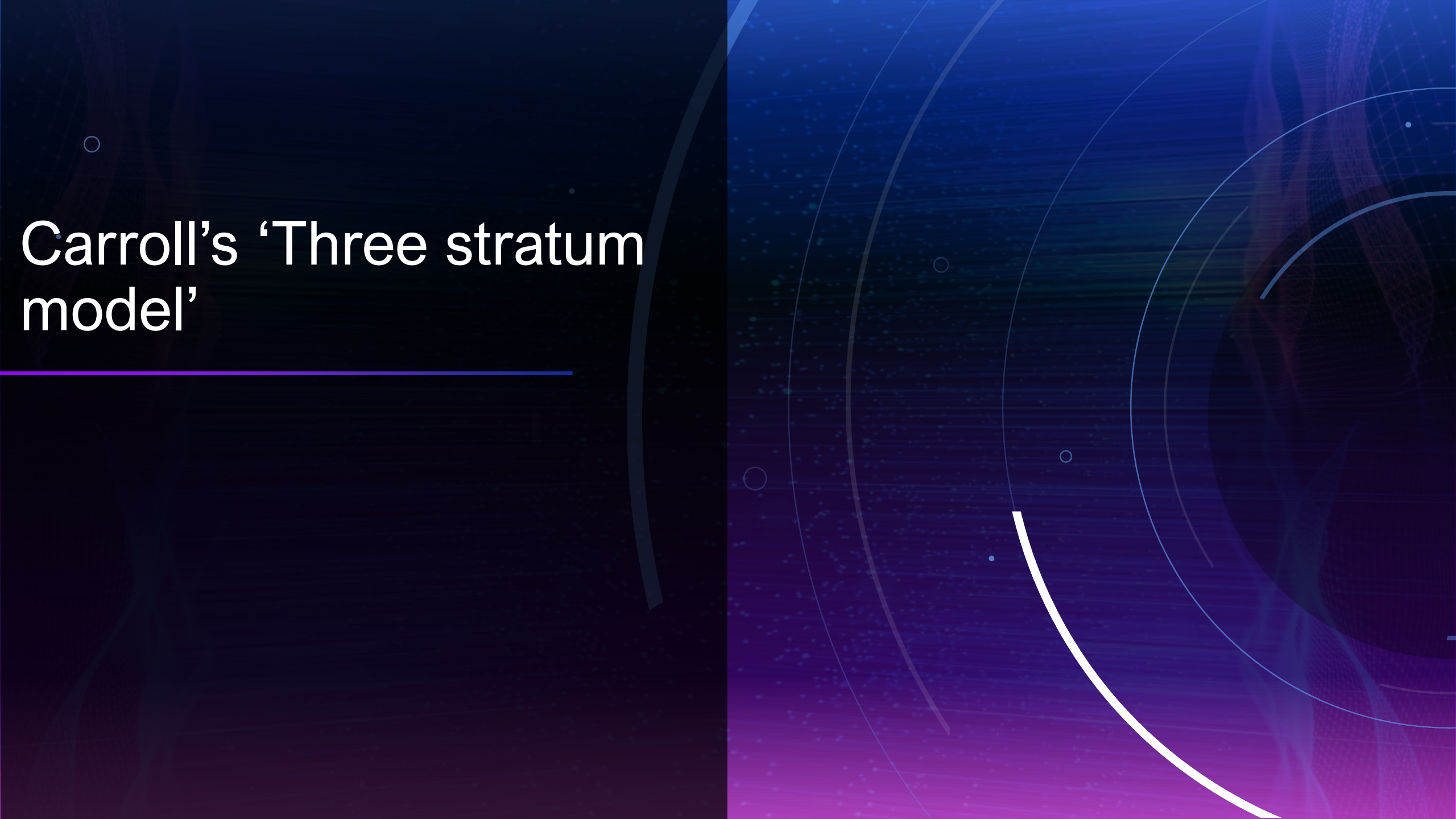
Thurstone's bottom-up approach



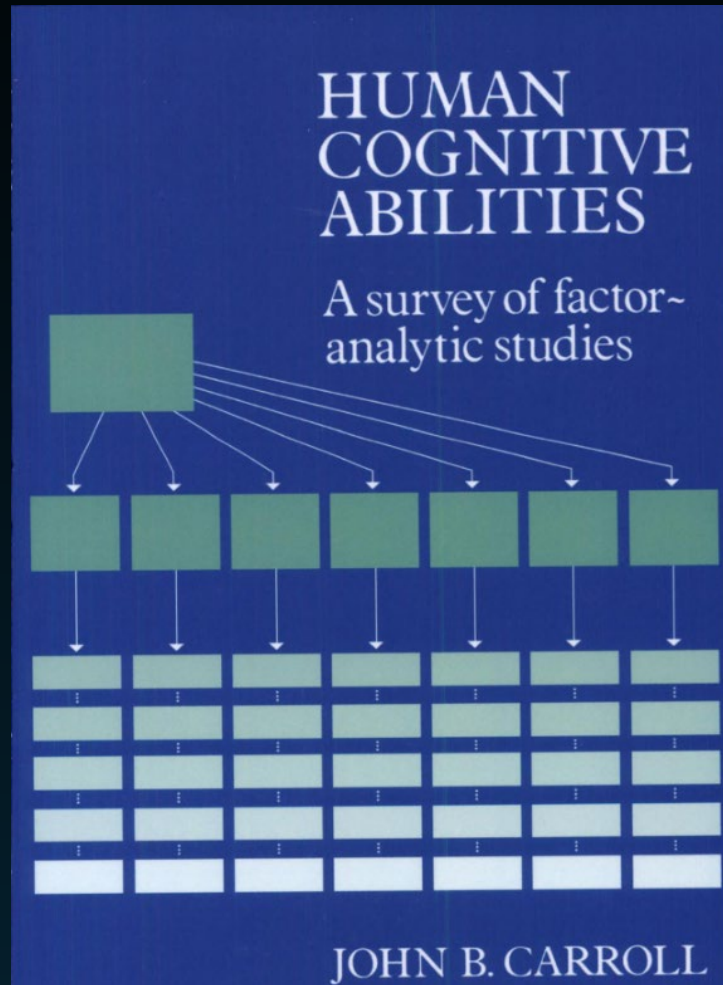
Emergence of mid-level abilities



Carroll's 'Three stratum model'

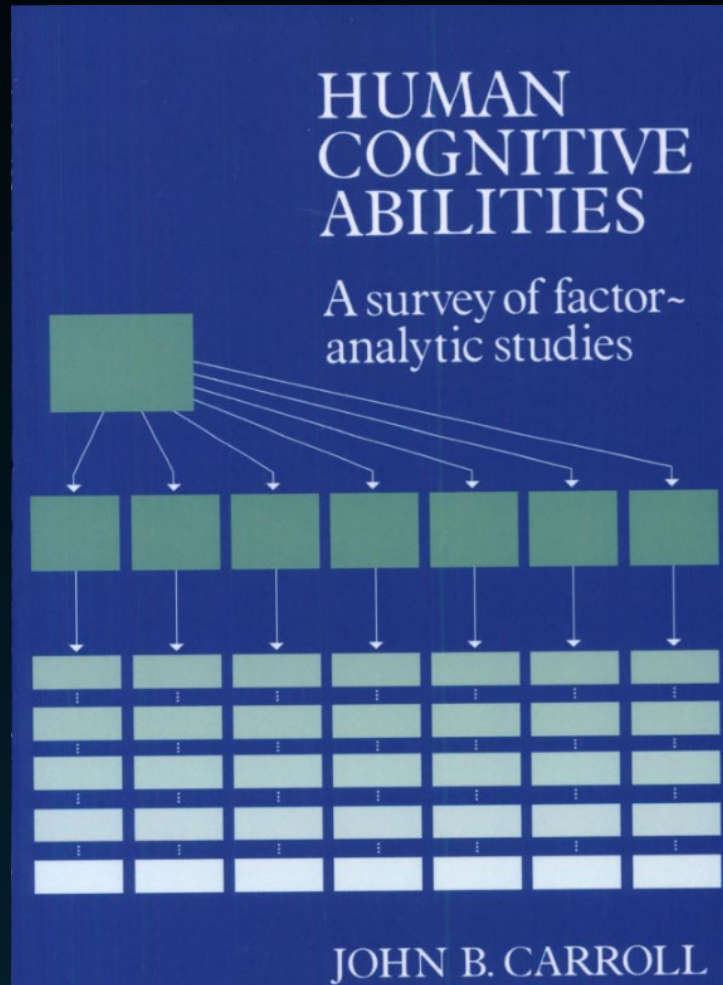


20th Century Spearman vs Thurstone debate brought to a close?



- John Carroll book (1993) 800 pages long! – mammoth task of reanalysing over 400 sets of reliable data on human mental ability testing
- 130,000 healthy adults, school children, infants, university students, people with learning disabilities and people with mental and physical illness
- His analysis was summarised in a diagram he called the ‘three stratum model’ of human cognitive ability

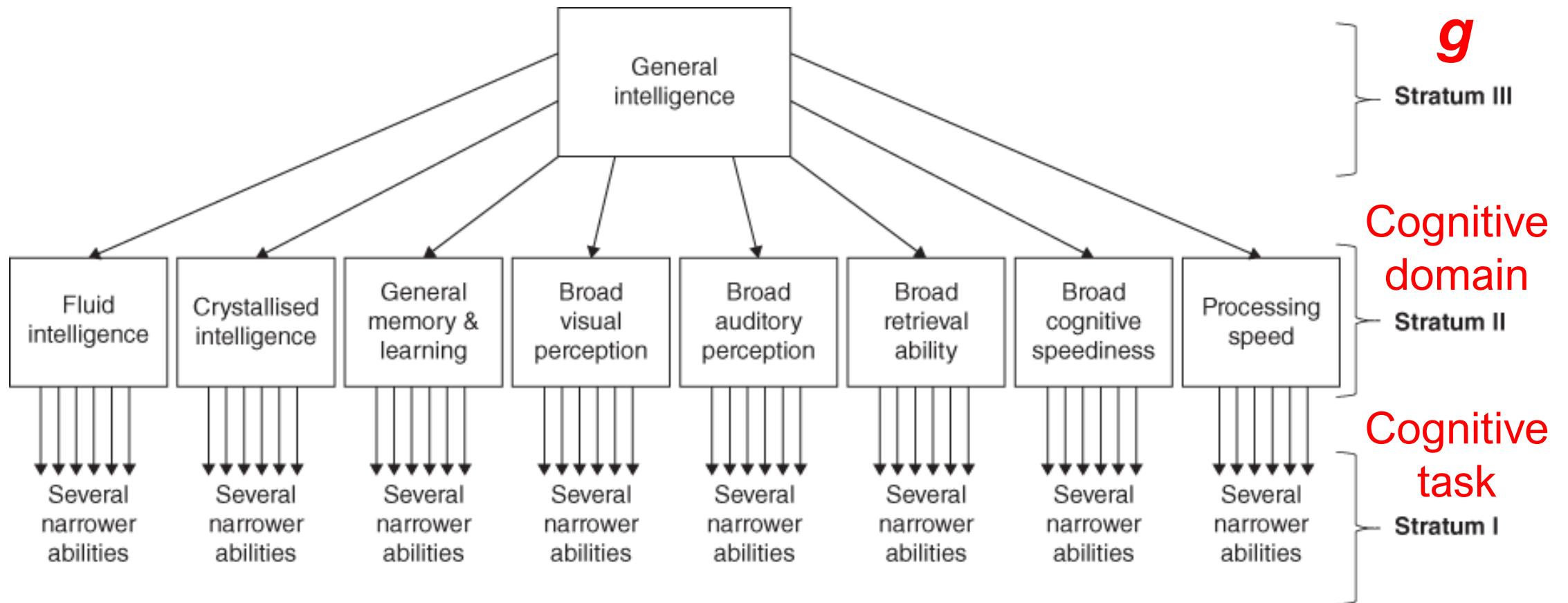
20th Century Spearman vs Thurstone debate brought to a close?



- John Carroll book (1993) 800 pages long! – mammoth task of reanalysing over 400 years of reliable data on human mental abilities
- 130,000 healthy adults, school children, infants, university students, people with learning disabilities and people with mental and physical illness
- His analysis was summarised in a diagram he called the 'three stratum model' of human cognitive ability

Both Spearman and Thurstone were kind of right!

‘Three stratum model’ of human cognitive ability



Bottom line: People obtain a good score for at least four possible & non-exclusive reasons:

- They are good overall (have high general intelligence)
- They are good at that sort of test (high ability for that cognitive domain)
- They are good at that specific test (high ability for specific cognitive task)
- Chance favoured them on that particular testing day

Cattell's *Fluid* versus
Crystallized intelligence



Raymond Cattell
1905-1998

Journal of Educational Psychology
1963, Vol. 54, No. 1, 1-22

Key article >4911 citations (as of Jan 2025)

THEORY OF FLUID AND CRYSTALLIZED
INTELLIGENCE:
A CRITICAL EXPERIMENT¹

RAYMOND B. CATTELL
University of Illinois

General intelligence (g) can be subdivided into:

- **Fluid** (g_f) intelligence and
- **Crystallized** (g_c) intelligence

Fluid (g_f) intelligence

Primary reasoning ability

- Abstract
- Not cultural
- Better tested by Raven's matrices

e.g. inductive reasoning in abstract sequences

Crystallized (g_c) intelligence

Acquired knowledge/skills

- Related to cultural/historical background of individuals
- Better tested by Weschler scales

e.g. vocabulary, social rules

Fluid (g_f) intelligence

“Fluid intelligence refers to reasoning and novel problem-solving ability. It is distinct from crystallised intelligence, which refers to overlearned skills and static knowledge such as vocabulary. Empirically, G_f is strongly associated with working memory”

Gray and Thompson (2004)
Nature Reviews Neuroscience

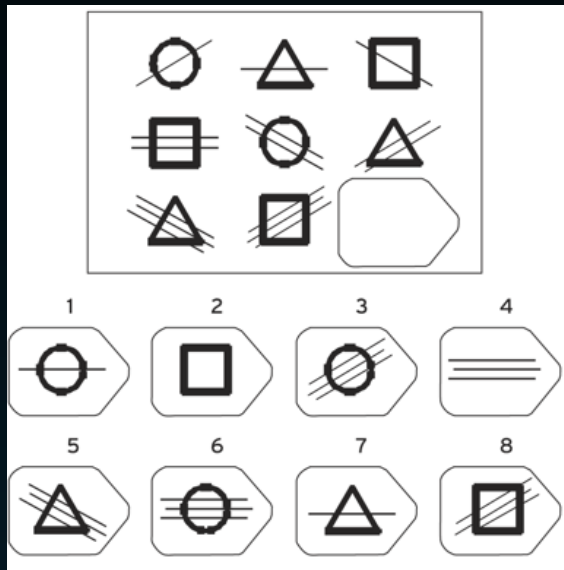
Crystallized (g_c) intelligence

“Crystallised intelligence is acquired knowledge and skills, such as factual knowledge. It is generally related to a person’s stored information and to their cultural influences...Crystallised intelligence (for example knowledge) is intelligence that increases throughout life and is a reflection of one’s cumulative learning experience.”

Maltby et al, page 294

Fluid (g_f) intelligence

Raven's Matrices better

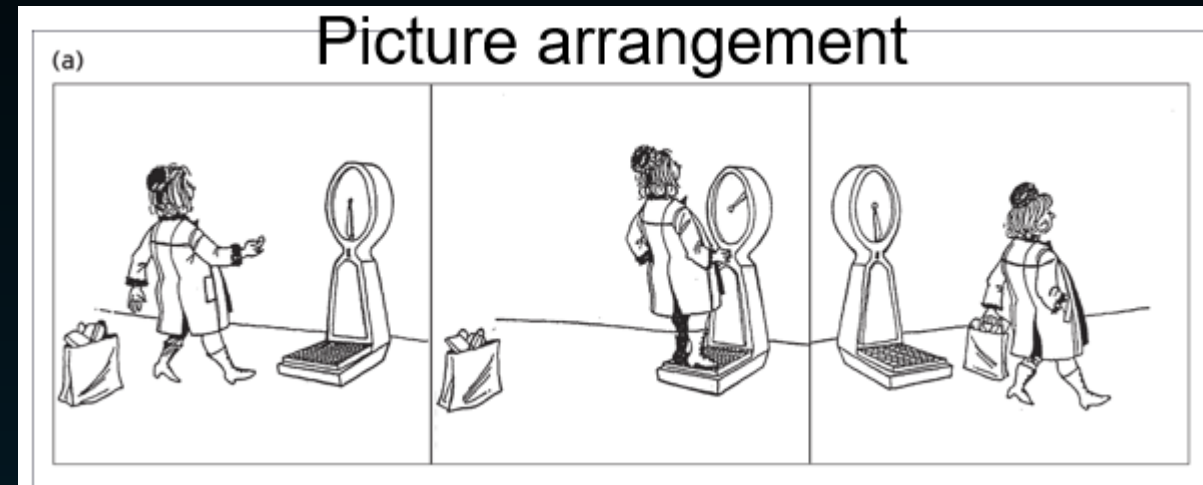


Complete the missing pattern

“Certainly Raven’s best measure of g_F , fluid intelligence.” Jensen (1988)

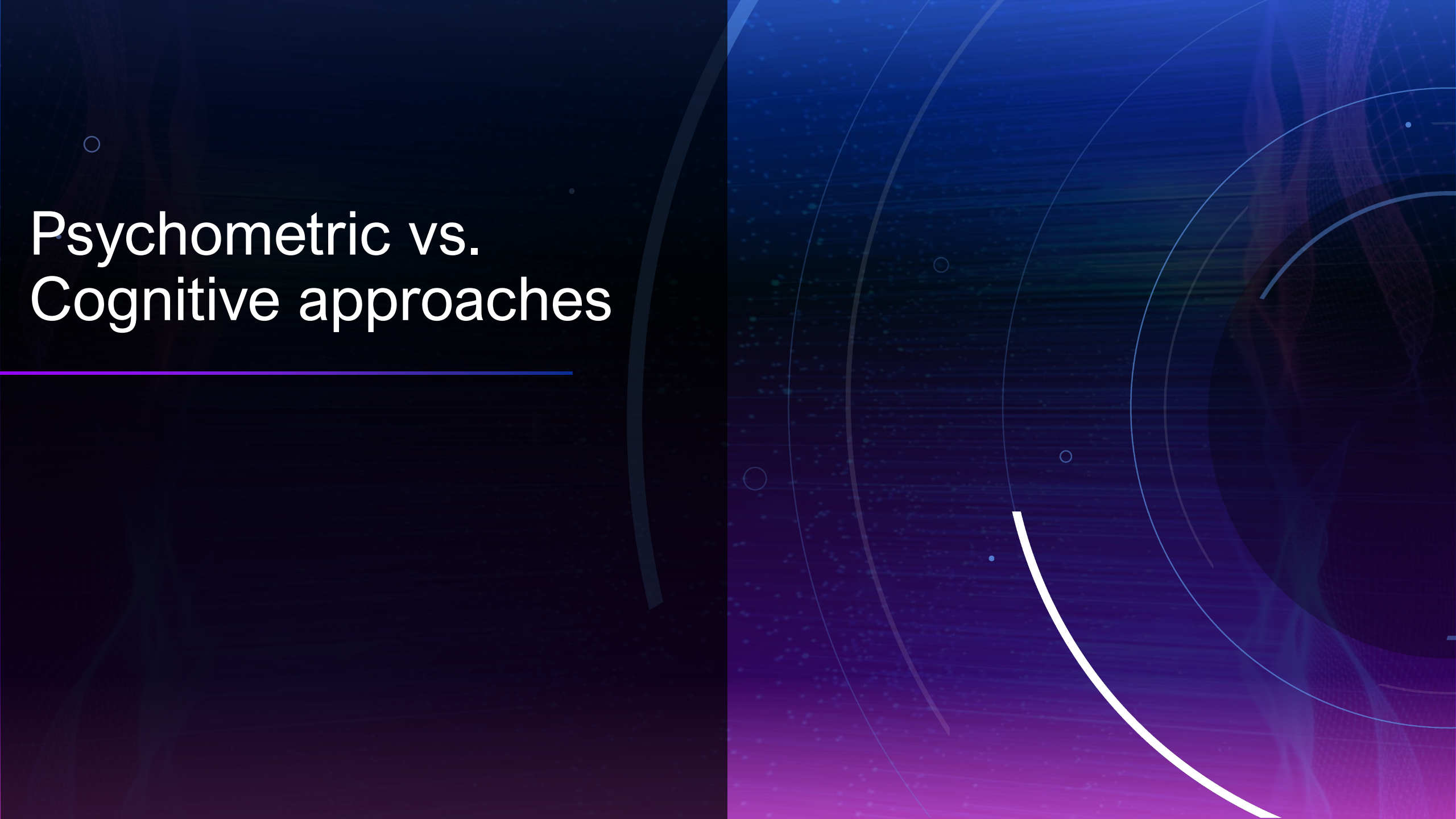
Crystallized (g_c) intelligence

Wechsler Tests better



Arrange pictures in correct sequence

Crucial to know culture bound fact like the vertical clock like device is a weighing machine



Psychometric vs. Cognitive approaches

Psychometric vs. Cognitive approaches

Psychometric Approach

Based on the findings of factor-analytic studies that have looked at the various psychometric properties of intelligence scales. Describes how people tend to differ but it cannot explain *why*. Focused on *whether* people answer items correctly

“What are the statistical patterns in test scores?”

Cognitive Approach

The cognitive approach is based on another set of intelligence tests that highlight biological and physiological processes involved in intelligence (lab based). Focused on how people answer the items, and why some people are better than others at various mental abilities.

“What are the mental processes underlying task performance?”

Psychometric vs. Cognitive approaches

Psychometric Approach

Based on the findings of factor-analytic studies that have looked at the various psychometric properties of intelligence scales. Describes how people tend to differ but it cannot explain *why*. Focused on *whether* people answer items correctly

Cognitive Approach

The cognitive approach is based on another set of intelligence tests that highlight the physical processes of the mind. Underlying cognitive information-processing. Like a computer - efficiency of problem-solving will be partly dependent on the limits of memory storage and speed of processing.

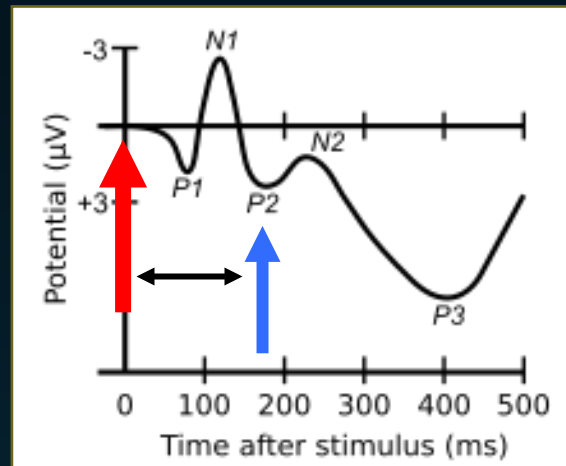
Simple Biological and Physiological Measures of Intelligence

Elementary Cognitive Tasks and Intelligence

- Elementary cognitive tasks (ECTs) are simple tasks to measure cognitive processes, such as understanding stimulus, stimulus discrimination, visual search, retrieval of information / perceptual speed, etc.
- Measuring response time:
 - Median reaction time (average response time over a number of trials)
 - Standard deviation of reaction time (RTSD) (individual variability in response time over a number of trials)
 - Inspection time (IT) / Evoked potential (EP) – the time people take to process (visual or auditory) information.



e.g. Barret & Eysenk (1994)



A shorter delay between the stimulus (red arrow) and EEG response (blue arrow) is thought to represent higher intelligence.

Simple Biological and Physiological Measures of Intelligence

→ Elementary Cognitive Tasks and Intelligence



- Elementary cognitive tasks (ECTs) are correlated with traditional measures of intelligence, the correlations averaging about $r = .35$.
- Overall scores based on the use of various ECTs with different measures of response times are found to correlate with scores on general intelligence tests of between $r = .50$ and $r = .70$.
- ECTs are useful because they involve no past learned information.

Multiple Intelligences Approach



Howard Gardner

- Not psychometric and not cognitive in the traditional sense.
- Gardner argued that traditional IQ tests measure only a narrow slice of human abilities (mainly linguistic and logical-mathematical)
- Proposed that intelligence is **plural**, not singular — humans possess **multiple independent intelligences**, each with its own developmental trajectory and neural basis.
- Emphasis on real-world functioning, cultural value, and adaptive usefulness of different skills.
- Intelligences are partly modular, meaning that damage to one brain system may impair one ability but leave others intact (e.g., musical talent preserved despite language deficits).

Gardner rejected the idea of a single “g” factor, arguing instead that humans have several relatively independent intelligences, each rooted in different brain systems and serving different adaptive functions.

Multiple Intelligences Approach

Gardner's theory grew from the following influences:

Neuropsychological dissociations

Selective brain damage can impair one ability (e.g., language) while sparing others (e.g., music, spatial skills)

Prodigies & Savants

Exceptional strengths in a single domain alongside profound weaknesses elsewhere show that abilities can develop independently.

Developmental trajectories:

Children show uneven skill growth (e.g., early musical talent but average reasoning), implying domain-specific development.

Cross-cultural observations:

Different societies value different adaptive skills (navigation, storytelling, social leadership), indicating multiple forms of competence.

Evolutionary considerations:

Proposed intelligences map onto capacities that likely offered survival advantages (e.g., spatial, interpersonal, naturalistic).

Symbol systems:

Each intelligence uses its own representational system (musical notation, language, maps, gestures), consistent with distinct cognitive domains.

Multiple Intelligences Approach



Howard Gardner



Frames of Mind: The Theory of Multiple Intelligences (1983)

Does Empirical Evidence Support the MI Theory?



Howard Gardner

There is evidence for:

- Neuropsychological dissociations (e.g., amusia vs preserved language, spatial impairments but intact social intelligence).
- Domain-specific learning systems in cognitive science (e.g., numerical cognition, language acquisition, face processing).
- Individual differences profiles, where people genuinely show strengths in one domain but not others.

Major Criticisms / Limited Evidence

- Basis of Gardner's model is not empirical. Like Freud, he provides more of a literary analysis of insight. Despite conceptual appeal, MI lacks strong factor-analytic / psychometric or other quantitative approaches.
- Gardner's intelligences are not statistically independent: his proposed intelligences tend to correlate with each other. Factor analyses consistently produce a large *g* factor, plus a few broad group factors (verbal, spatial, memory, etc.)
- **However, very popular theory beyond academia** > appeals to ideal we all have some special ability, > doesn't recognize limitations implied by *g*, > based on anecdotes / observations easy to understand

SUMMARY

- Factor analysis - statistical technique to represent correlations between intelligence tests, which can be clustered into more specific factors, e.g. reasoning, vocabulary etc.
- Correlations between all factors are positive (a highly replicated result) – implying that all the factors derived from individual tests have something in common - called the **g-factor** or **g**
- Spearman's 2-factor theory of intelligence: *specific abilities* ('s') and general ability ('g')
- Thurstone, suggested that there were 7 separable human mental abilities.
- John Carroll's 'three stratum model' of human cognitive ability allows for individual variability in cognitive domains between individuals with similar IQ scores
- Cattell subdivided **g** into *Fluid* and *Crystallized* intelligence
- Useful to consider cognitive processes as well as statistical patterns in IQ scores
- Reaction times / processing speed fairly well correlated with **g**
- Gardner's popular model proposes multiple intelligences but borne out of observation more than quantitative approaches.

END